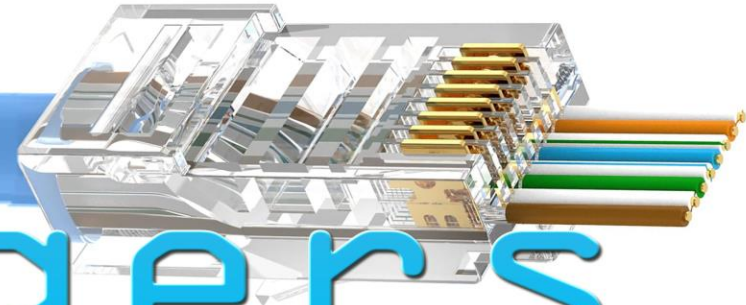


Modbus

Messengers



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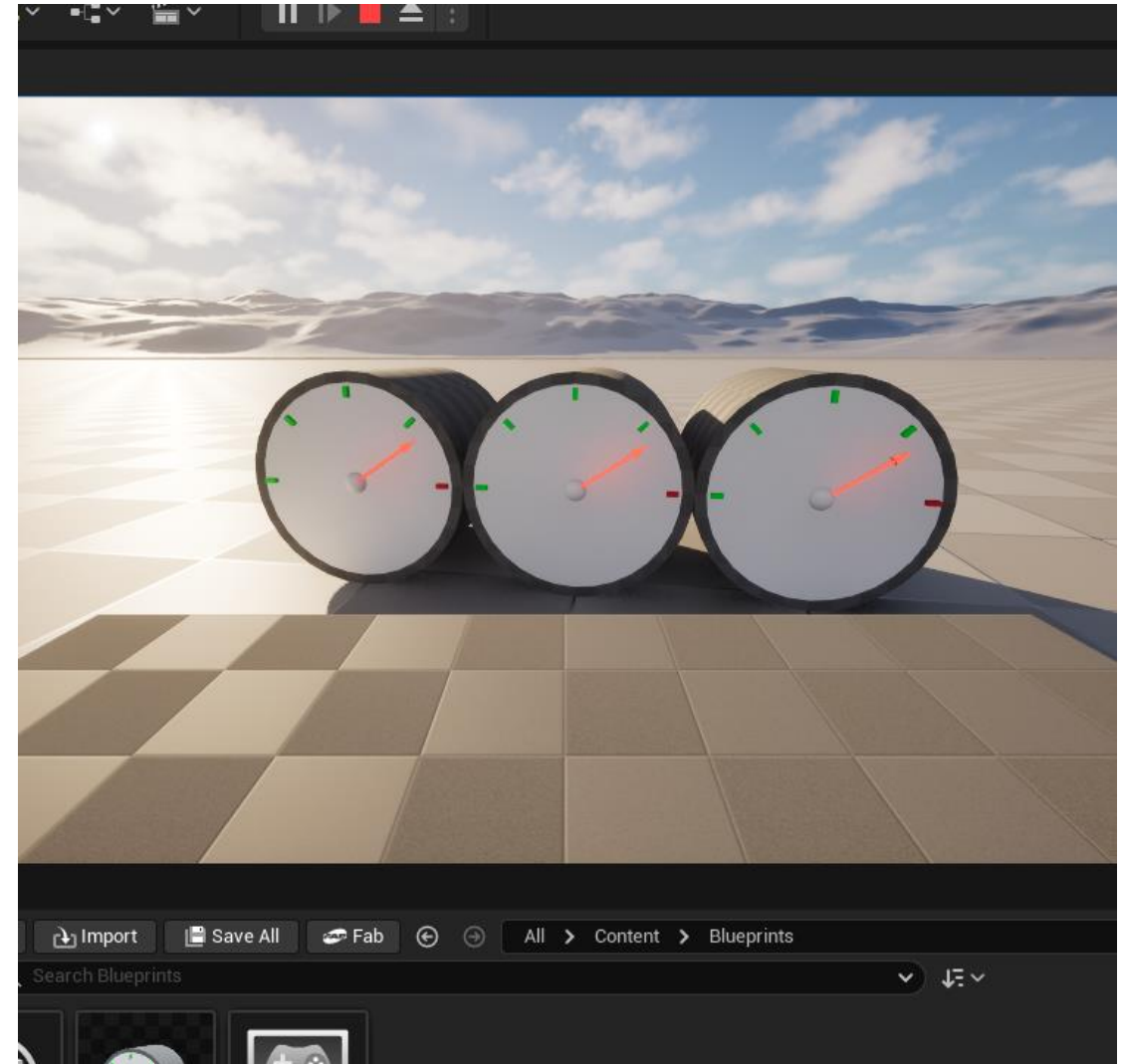


Project 47 Scope – Integration of Modbus Functionality into Unreal Engine

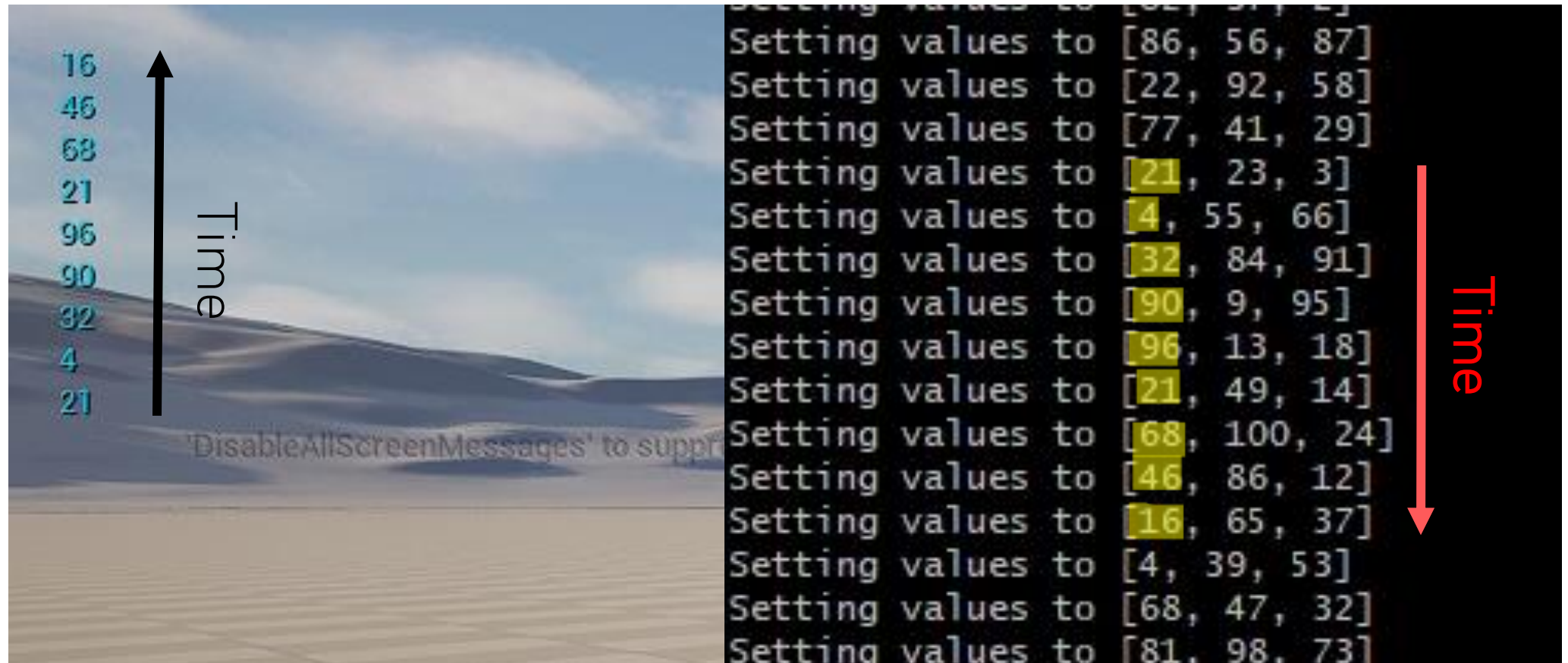
- Create a plugin for Unreal Engine that will communicate asynchronously with a real-world endpoint device and simulated digital twins built in the Unreal Engine environment via Transmission Control Protocol (TCP).
- An example will need to be provided, displaying that a change originating from the device shows a change on the digital twin and vice versa.

Progress Update

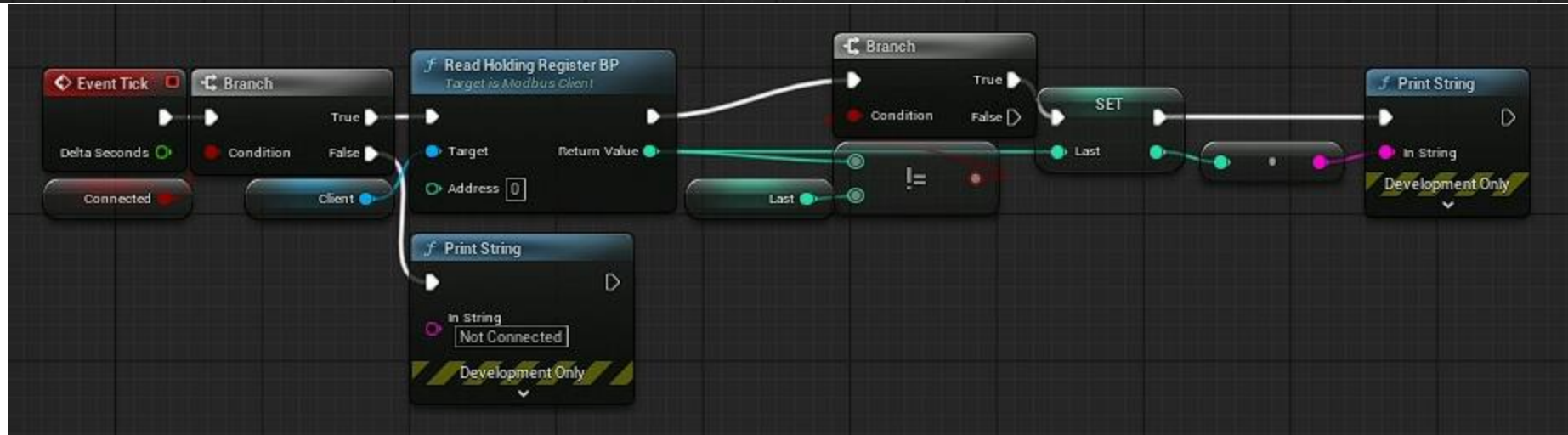
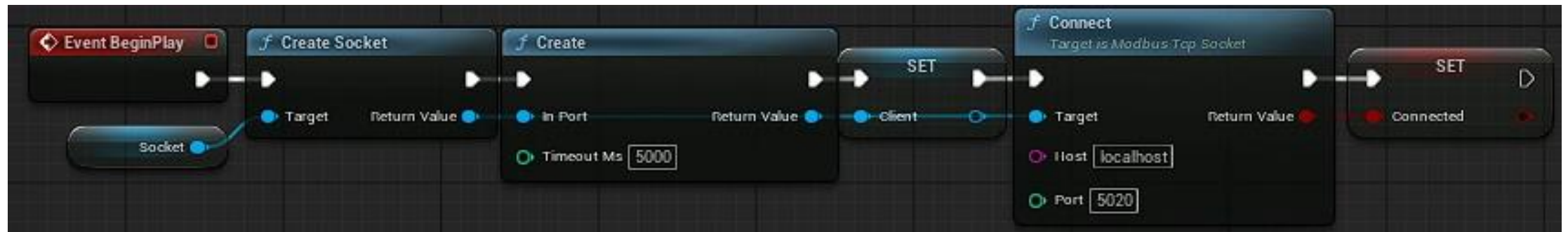
- Emulator
 - Scripts to start emulator and read from simulated device created.
 - Modbus encoded data being successfully transmitted over opened port.
- Plugin
 - nanoModbus library integrated and in use.
- Sample Scene
 - Receiving Modbus encoded data from the emulator server.
 - Basic valves representing data from emulator using rotation of needles.



Progress Update – UE Reading Modbus Data



Progress Update – Blueprints (Connect & Read)



What's Next

- Expand on functionality within the Unreal Editor
- Add error handling
- Further develop and document the emulator to include specific device simulation
- Smooth out movement and stabilize the emulator outputs
- Expand on sample scene to better represent potential of project



Current Hurdles

- Conflicts when working within Unreal Engine
 - Maintaining uniform Blueprint visibility
 - Keeping nanoModbus integrated – Unreal does not use C code and has difficulty finding the header file location.
 - Issues with compiling the Unreal Engine project on Linux
 - Expanding on read capabilities from the server





Sustainability

Utilization of Modbus is complex and time-consuming, often requiring custom solutions for each project.

The goal of this project is to utilize the ability offered by the Unreal Engine Blueprints system to create a plugin that allows for the control of real-world Modbus devices.

This will allow for faster and safer creation of digital twins for Modbus systems using the Unreal Engine environment and provide a cost-saving, generic interface for controlling Modbus devices to a wide range of developers.



Validation Plan

Requirements	Test	Subject	[Goal] Completion Date
Integrate Modbus to Unreal plugin	Responsive code through sample digital twin	Plugin code	10/21/2025
Digital twin can use Modbus	Responsive sample digital twin in Unreal environment	Sample digital twin	12/9/2025
Unreal plugin implements Blueprints	Modbus interactions created through Blueprints system	Blueprint plugin	3/31/2026
Plugin communicates with devices on TCP using Modbus	External device is controllable through Unreal interface	External device such as PLC	4/24/2026

Software Unit Tests Not Included



What is Modbus?

- Widely used open-source communication protocol for electronic devices
- Industrial control systems in factories and power grids
- Used in green energy solutions such as wind turbines and solar farms
- Master / Slave setup. Slaves (Sensors, PLCs, etc.) have four kinds of registers:

Register Type	Read/Write	Data Type	Used For
Coils	R/W	1-bit	Digital outputs
Discrete Inputs	Read-only	1-bit	Digital inputs
Input Registers	Read-only	16-bit	Sensor values, analog input data
Holding Registers	R/W	16-bit	Config, analog output, sensor data



What Software Are We Using?

- Unreal Engine v5.6.1
 - Necessary; see project requirements.
- nanoMODBUS
 - Lightweight, MIT licensed C library to provide Modbus support in C and C++ (UE's backend language).
- PyModbus
 - Comprehensive Modbus simulator written in Python.